Monitoring

This guide covers monitoring the health and performance of the KUCCPS IT Ticketing System.

Health Check

Endpoint

```
GET /health
```

Response

```
{
  "status": "OK",
  "message": "KUCCPS Ticketing System API is running"
}
```

Monitoring Tools

Use external monitoring services to check the health endpoint:

Tool	Setup
UptimeRobot	Free HTTP monitor
Pingdom	Advanced monitoring with alerts
Datadog	Full-stack observability
Grafana + Prometheus	Self-hosted metrics

Application Metrics

Key Metrics to Track

Metric	Description	Alert Threshold
API response time	Average request latency	> 500ms
Error rate	5xx responses per minute	> 5
Ticket creation rate	Tickets created per hour	Anomaly detection
Active users	Concurrent authenticated users	N/A
Database connections	Active DB connections	> 80% of pool
Disk usage	Upload directory size	> 80%

Logging

Log important events with request IDs:

```
console.error('Unhandled request error:', { requestId: req.id, message: err.message });
```

Log format should include:

- Timestamp
- Request ID
- Log level (info, warn, error)
- Message
- Context (user ID, ticket ID, etc.)

Database Monitoring

Connection Count

```
SELECT count(*) FROM pg_stat_activity WHERE datname = 'kuccps_ticketing';
```

Slow Queries

Enable slow query logging in `postgresql.conf`:

```
log_min_duration_statement = 1000  # Log queries > 1s
```

Table Size

```
SELECT
    schemaname,
    tablename,
    pg_size_pretty(pg_total_relation_size(schemaname, tablename)) AS total_size
FROM pg_tables
WHERE schemaname = 'public'
ORDER BY pg_total_relation_size(schemaname, tablename) DESC;
```

Email Monitoring

Monitor email delivery:

- Check SMTP logs for bounces and failures
- Monitor IMAP connection health
- Track email queue depth
- Alert on consecutive failed sends

Alerting

Set up alerts for:

1. **Critical:** Health check failure, database down, error rate spike
2. **Warning:** High response time, approaching disk capacity, failed email sends
3. **Info:** Daily digest sent, backup completed

Alert Channels

- Email to administrators
- Slack/Discord webhook
- SMS for critical alerts
- PagerDuty for on-call escalation

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Deployment Guide

Troubleshooting

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